🏛️ IIT Bombay — IDC

Electric Flute

Music You Can See ology You Can Play

An Arduino-powered flute that lights up LEDs for each musical note (Sa Re Ga Ma Pa Da Ni Sa) — built at IIT Bombay's Industry Design Centre

Arduino | Sound Sensor | WS2812B LEDs | Piezo | Bamboo Lab 3D Print

For Students: Class 1 – 12 | A STEM Toy Design Project

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PROJECT OVERVIEW

What is the Electric Flute?

Built at IDC — IIT Bombay

The Big Idea 💡

A real bamboo flute transformed into a smart electronic instrument. Every note you blow — Sa, Re, Ga, Ma, Pa, Da, Ni — lights up a corresponding LED in real time!

🎵 Real Flute Body

Actual bamboo flute with LEDs attached to note holes

🔊 Sound Detection

Microphone sensor picks up frequency from each note

💡 LED Visualizer

WS2812B addressable LEDs glow per Sargam note

🧠 Brain: Arduino

Arduino Uno processes sound and controls all LEDs

🎹 Piezo Sensor

Detects physical vibrations to identify precise Sa–Ni notes

🖨️ 3D Printed Parts

Custom mounts printed with Bambu Lab P1S printer

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ELECTRONICS BASICS — The Building Blocks

🧠 The Brain

Arduino UNO (Microcontroller)

- MCU = Micro-Controller Unit
- Has 14 digital I/O pins
- Runs code written in C/C++
- Reads sensor signals
- Controls LEDs via PWM
- Clock Speed: 16 MHz
- Programmed via Arduino IDE

💡 The Eyes

WS2812B LEDs (Addressable LEDs)

- Each LED is individually addressed
- 1 wire controls colour + brightness
- Red + Green + Blue (RGB) mixing
- Each note = unique LED colour
- Uses NeoPixel / FastLED library
- Can glow 16 million colours
- Driven by 5V DC power

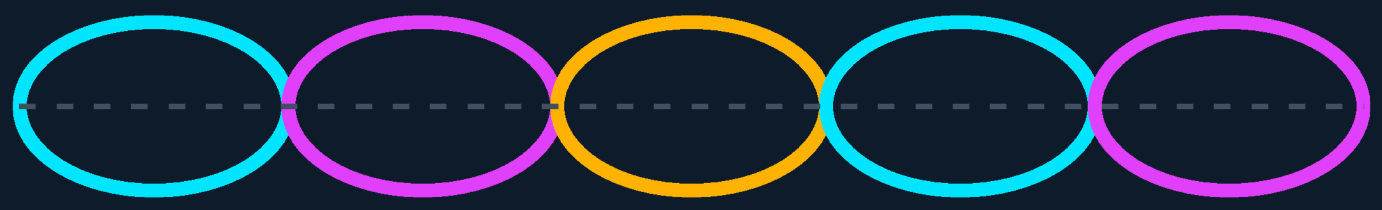
🔊 The Ears

Sound & Piezo Sensors

- Microphone detects air pressure waves
- Converts sound → voltage signal
- Arduino reads analog values (0-1023)
- Piezo detects vibration frequency (Hz)
- Sa=261Hz, Re=293Hz, Ga=329Hz…
- Filters noise to identify correct note
- Analog Input Pin (A0, A1) used

THE PHYSICS OF SOUND — Waves, Frequency & Notes

Sound = Waves of Air Pressure 🌊



Wavelength →

Frequency = How many waves per second (Hz)

Sa (C4)	261 Hz
Re (D4)	293 Hz
Ga (E4)	329 Hz
Ma (F4)	349 Hz
Pa (G4)	392 Hz
Da (A4)	440 Hz
Ni (B4)	493 Hz
Sa' (C5)	523 Hz

What is Frequency?

Frequency = number of wave cycles per second.
Unit: Hertz (Hz). Higher Hz = higher pitch.
Human hearing: 20 Hz – 20,000 Hz

How a Flute Makes Sound

Blowing across the hole creates a vibrating air column. Covering different holes changes the column length, changing the frequency = different notes!

Amplitude = Loudness

The height of the wave = how loud it is.
Sensor reads this as voltage (0 to 5V).
Arduino converts it to 0–1023 (10-bit ADC).

Digital vs Analog

Sound is analog (smooth wave).
Arduino's ADC converts it to digital numbers.
This lets us detect which note is being played!

05 HOW IT WORKS — System Flow & Circuit Logic

Player Blows Flute — Mic + Piezo Sensor — Analog Signal (Hz) — Arduino Processes — LED Lights Up (Note) — Visual Feedback

1 Blow the Flute

Air passes through the flute. Covering finger holes changes the vibrating air column length, producing different musical notes (Sa–Ni).

2 Sensor Reads Sound

The microphone (sound sensor) picks up the sound wave. The Piezo sensor senses physical vibrations in the flute body and sends analog voltage to Arduino.

3 Arduino Identifies Note

Arduino reads the analog signal (0–1023). Using frequency thresholds programmed in Arduino IDE, it identifies which note (Sa/Re/Ga...) is being played.

4 LED Signal Sent

Arduino sends a digital signal to the WS2812B LED strip using a single data wire. Each note maps to a specific LED index and colour in the strip.

5 LED Glows!

The correct LED lights up in a unique colour — Sa=Blue, Re=Green, Ga=Yellow, Ma=Orange... making music visible on the flute body!

6 Real-Time Visual

The whole process happens in milliseconds. As you play different notes one after another, LEDs switch in real time — music becomes a light show!

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WHAT KIDS LEARN — Grade-wise Learning Outcomes

This project bridges Music + Physics + Electronics + Coding — for every age group!

🌱 Class 1–4

- What is sound?
- How do musical notes differ?
- Colours of LEDs (RGB)
- What does a battery do?
- Fun: Match note to colour!

🔬 Class 5–7

- Sound waves & vibration
- Frequency vs pitch (Hz)
- Conductors & insulators
- Simple circuits & LEDs
- How sensors work

⚡ Class 8–10

- Analog vs Digital signals
- ADC (Analog-to-Digital)
- Resistors, capacitors
- Ohm's Law in LED circuits
- Introduction to Arduino

🚀 Class 11–12

- Microcontroller architecture
- C/C++ Arduino programming
- PWM & signal processing
- Frequency detection algorithms
- PCB design & IoT concepts

07 BUILD IT YOURSELF — Step-by-Step Guide for Students

🛒 What You Need

- Arduino Uno R3
- WS2812B LED Strip (8 LEDs)
- KY-038 Sound Sensor
- Piezo Transducer Disc
- Bamboo Flute (real)
- USB Cable + 5V Adapter
- Jumper Wires + Breadboard
- Bambu Lab P1S (3D Printer)
- Arduino IDE (free software)

🔧 Build Steps

1 Design & 3D Print Mounts

Use Bambu Lab P1S to 3D print LED holders and sensor clips that fit your flute's body.

2 Wire the Sound Sensor

Connect KY-038 microphone to Arduino A0 pin. VCC to 5V, GND to GND.

3 Connect Piezo Sensor

Tape the Piezo disc to the flute body. Connect to A1 pin through a 1MΩ resistor.

4 Set Up LED Strip

Connect WS2812B data pin to Arduino D6. Power via 5V. Install FastLED library.

5 Upload the Code

Write/upload Arduino sketch: read sensor → detect Hz → map note → light LED.

6 Test & Calibrate

Play each note. Adjust frequency thresholds in code until each Sa–Ni lights correct LED.

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FUTURE SCOPE & YOUR TURN TO BUILD!

🔭 What Can Be Added Next?

📱 Bluetooth App

Send note data to a smartphone app and display real-time musical score!

🖥️ OLED Screen

Add a tiny screen to display note name, frequency, and wave visually.

🎵 Auto-Tune Mode

Arduino detects if note is slightly off and shows warning LED.

🌐 IoT Dashboard

Log all notes played to a cloud dashboard using Wi-Fi (ESP32 upgrade).

🤖 AI Note Trainer

Machine learning model to recognize patterns and suggest next notes.

🎮 Music Game

LED game: correct note = points! Makes learning Ragas into a game.

🚀 YOUR TURN!

- ✅ Identify 3 electronic components at home
- ✅ Download Arduino IDE — it's FREE!
- ✅ Blink an LED with 5 lines of code
- ✅ Play Sa on a flute & measure Hz
- ✅ Map a note to an LED colour
- ✅ Present your version of this project!

"Every great engineer started by asking: What if I tried building it?" — IIT Bombay IDC